My Only Chronic Condition is Acing Stats!!

Determining How Patients with Different Chronic Conditions Respond to Nucleus' Services.

Brian Xiao, Jiani Yang, Johnny Li | Group 0114-E

2025-11-28

Introduction

Our project examines how clients with different chronic conditions respond to Nucleus services, focusing on how their healthcare use changes after enrollment.

Using a sample of roughly 4,000 clients, we aim to generalize findings to the broader Nucleus population.

The analysis is intended for Nucleus program leaders and care teams to better tailor to support clients with higher needs.

Question 1: Jiani Yang – Hospital Days VS Chronic Burden

Among clients with chronic conditions, how does the *typical* change (median) in hospital days differ across groups with 0, 1, 2, 3, 4, or 5+ chronic conditions?

- Clients with multiple chronic conditions often have higher hospital use.
- Understanding which burden groups reduces hospital days help Nucleus identify which clients benefit most from the program

Variables & Approach

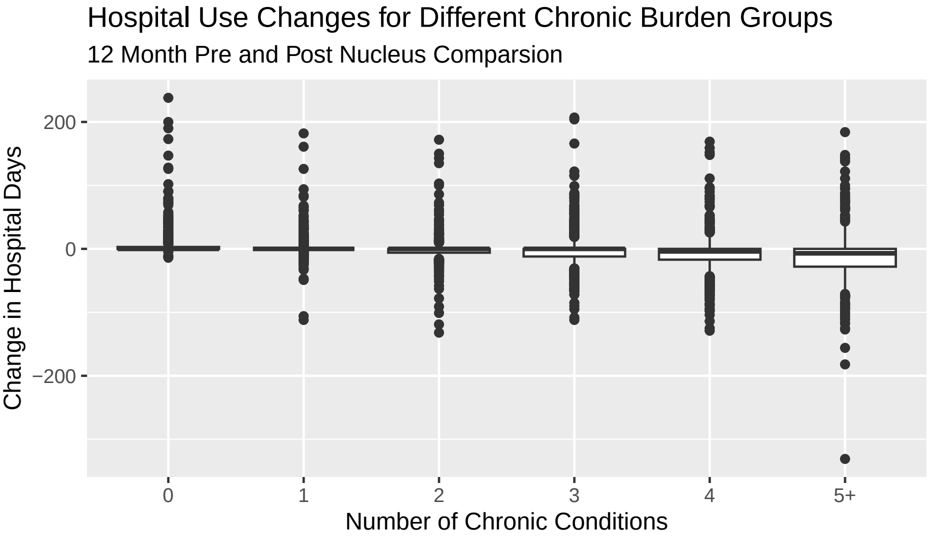
Key Variables:

- Chronic burden group
  - Groups clients by number of chronic conditions (0, 1, 2, 3, 4, 5+).
- Change in hospital days
  - How many more or less days a client spent in the hospital 12 months after joining Nucleus compared to the 12 months before

Data Wrangling — New variables/groups defined:

- Removed observations missing chronic condition and hospital use records
- Counted number of chronic conditions and categorized into burden groups of 1, 2, 3, 4, or 5+ conditions
- Calculated change in hospital days
- Restricted change in days to 12 month range

Q1: Visualization



Q1: Summary Table

Table 1: Comparison of Change in Hospital Stay Days Based on Chronic Condition Group

Chronic Group	Count	Mean Change	Median Change
0	497	8.09	0
1	585	3.32	0
2	744	-0.79	0
3	795	-2.39	0
4	635	-6.84	-4
5+	743	-10.95	-7

Q1 Insights:

- The mean is influenced by outliers, which makes the average look higher or lower for some groups
- Consider the median for the *typical* client:
  - 0, 1, 2, and 3 chronic conditions: **0 days (**no meaningful increase or decrease**)**
  - 4 chronic conditions: **4 day reduction**
  - 5+ chronic conditions: **7 day reduction**
- Results suggest Nucleus may have the strongest impact on reducing hospital days for clients with more chronic conditions.

Q1 Limitations:

Limitations:

- One client had an impossible value of 477 days within a 12-month range due to overlapping hospital stay records. This was removed for results to reflect realistic hospital use.
- Clients with missing data in variables were removed, which reduces size of data used to work with
- Chronic burden treats all conditions equally, though severity differs

Future Directions:

- Use updated data set with full records
- Group clients by baseline healthcare usage, provided the data, rather than just number of conditions
- Explore how each chronic condition independently respond to Nucleus Services

Question #2: Brian Xiao – Chronic Condition VS Total Reduction

Do members with certain chronic conditions see greater reduction in total healthcare costs from Nucleus services than those with other conditions?

- Focuses on the monetary benefits for members using Nucleus Independent Living
- Determines whether members with certain chronic conditions benefit more, financially, than others.
- Allows Nucleus to better understand for whom their services impact the most

Q2: Variables & Approach

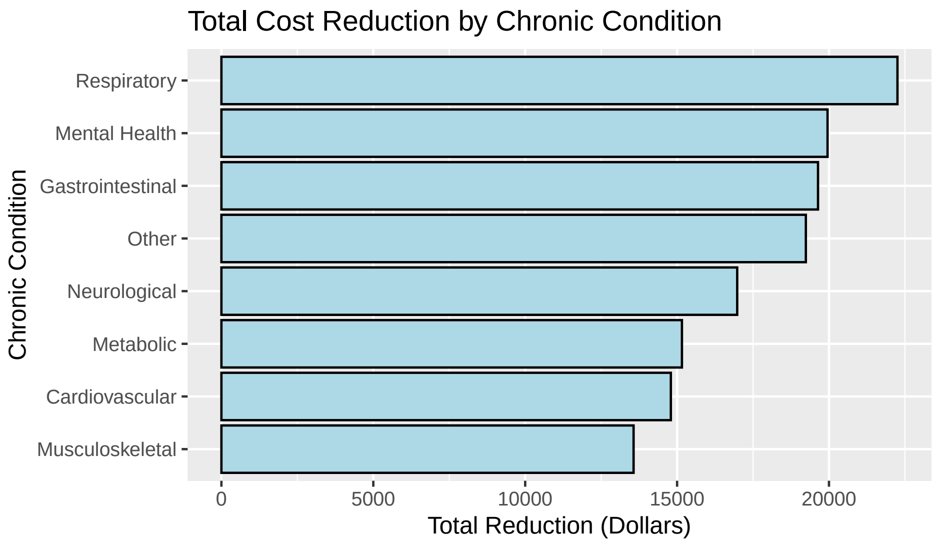
Key Variables:

- Certain chronic conditions
  - Patients were grouped based on which chronic condition they experienced.
- Reduction in cost
  - An average reduction in money spent towards hospital visits was calculated

Data Wrangling — New variables/groups defined:

- Created variable “Average reduction in cost” from different variables describing money spent before and after joining Nucleus.
- Created variable “Condition” that grouped patients by their chronic condition

Q2: Visualization



Q2: Summary Table

Table 2: Total Reduction in Cost by Chronic Condition

Condition	Count	Avg cost reduction
Cardiovascular	2561	14794.7
Gastrointestinal	1159	19638.9
Mental Health	1002	19953.4
Metabolic	1690	15159.9
Musculoskeletal	1729	13568.2
Neurological	1309	16978.8
Other	904	19238.9
Respiratory	811	22254.2

Q2 Insights:

- All chronic-condition groups show a meaningful reduction in healthcare costs after joining Nucleus, a positive indicator of program impact.
- Clients with respiratory conditions saw the largest average reduction (about $22,500), followed by those with mental health and gastrointestinal conditions (around $20,000).
- These results suggest that Nucleus support may be especially effective for clients whose conditions lead to avoidable or uncoordinated hospital use.

Q2 Limitations:

- Data counts people with multiple conditions as one data point for each condition, there may exist some confounding variables.
- Correlation between a group's chronic condition and their reduction in total cost may be attributed to other factors not accounted for in the data

Future Directions:

- Remove patients with multiple conditions to only observe independent ones
- Correlation is still important to study and should not be disregarded if causation cannot be proven.

Question #3: Johnny - Hospital Re-admits VS Chronic Status (across age groups)]

Among clients of different ages, do those with chronic conditions experience more unscheduled hospital admissions than clients without chronic conditions?

- Clients with chronic conditions might rely more heavily on hospitals and have higher hospital readmission rates
- Understanding how clients with and without chronic conditions and their readmission rates across age groups interact helps Nucleus see which groups need more support

Q3: Variables & Approach

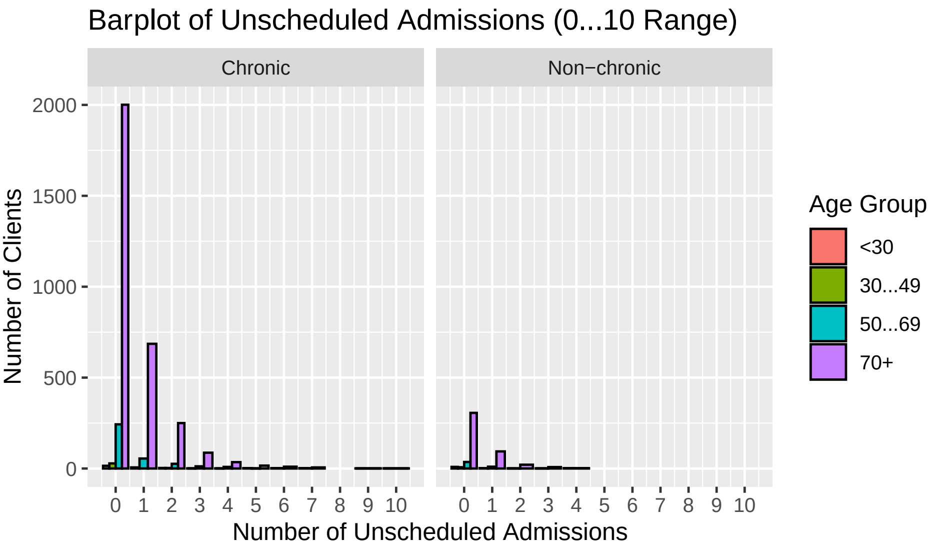
Key Variables Used:

- Unscheduled Admissions:
  - Number of unexpected overnight hospital stays in the 12 months after starting Nucleus.
- Chronic Status:
  - whether a client has at least one chronic condition or not
- Age Group:
  - Groups clients into 4 age groups of 10 year segments

Data Wrangling:

- Created a chronic vs non-chronic variable based on all condition indicators
- Grouped clients into 4 age categories
- Removed any clients missing age, chronic status, or hospital admission data.
- Restricted the graph to 0–10 admissions to avoid extreme outliers.

Q3: Visualization



Q3: Summary Table

Table 3: Average Number of Re-admits Based on Chronic Status Compared Across Age Groups

chronic_status	mean re-admit <30	mean re-admit 30-49	mean re-admit 50-69	mean re-admit 70+
Chronic	1	2.5	4.22	4.11
Non-chronic	0	0.5	1.50	2.00

Q3 Insights:

- Most clients, regardless of chronic status, had mean of 0 or 1 unscheduled hospital admission in the year after joining Nucleus.
- Clients aged 70+ have highest readmission counts, especially within the chronic group.
- Chronic clients have higher **mean** readmission counts than non-chronic clients across all age groups

Q3 Limitations:

- Age is grouped into 4 ranges, which simplifies graphs but does not show nuances within groups
- Chronic condition is treated as “yes” or “no” and disregards complexity of conditions
- Clients missing age, chronic status, or hospital admission data were removed from the analysis

Future Directions:

- Use more detailed age groups
- Consider the number and type of chronic conditions
- Use updated data set with full records

Overall Conclusion

Key Findings:

- Clients with higher chronic burden show the greatest reduction in hospital days.
- All chronic condition groups see reduced healthcare spending, especially respiratory and mental health conditions.
- Older clients with chronic conditions have the highest re-admission counts.

These results support the idea that Nucleus' services are especially impactful for clients with more complex chronic conditions, helping reduce hospital use and improving financial outcomes

Nucleus can continue monitoring patterns and prioritize care for clients with chronic profiles

References & Acknowledgements (if applicable)

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Acknowledgements:

- Thank you to Nucleus Independent Living for sharing the dataset for our educational purposes.
- Thank you to Professor Moon and Professor Griffith for cleaning the variables as well as supportive guidance
- Thank you to the STA130 TAs, specifically Yunfei, for providing constant feedback and help!